02/06/2026

Code-P

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **Project Phoenix for Re-NEET-2026_CST-04P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. There is a conducting rod of length 2*L* that lies along *x*-axis from $x = -L$ to $x = L$. If rod moves with velocity $v = v_0\hat{i}$ in the magnetic field $\vec{B} = B_0\hat{k}$, then induced emf between ends of rod is

(1) Zero

(2) $\frac{2}{3}B_0Lv_0$

(3) $\frac{1}{3}B_0Lv_0$

(4) $2\ B_0Lv_0$

2. **Statement A:** Action and reaction act on two different bodies.
Statement B: Action, reaction never cancel each other.

(1) A and B are correct

(2) A and B are incorrect

(3) A is correct and B is incorrect

(4) A is incorrect and B is correct

3. Magnetic susceptibility χ_m of a diamagnetic substance depend on absolute temperature *T* as $\chi_m \propto T^n$. The value of *n* is

(1) 1

(2) Zero

(3) –1

(4) –2

4. The amplitude of the magnetic field of an electromagnetic wave in vacuum is B_0 = 510 nT. The amplitude of the electric field in the wave is

(1) 120 N/C

(2) 134 N/C

(3) 510 N/C

(4) 153 N/C

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5. **Statement 1:** When battery acts like a source, the potential difference across its terminals is less than its emf.
Statement 2: When battery acts like a load, the potential difference across its terminals exceeds its emf.
Statement 3: When no current flows through the battery, the potential difference across its terminals is zero.

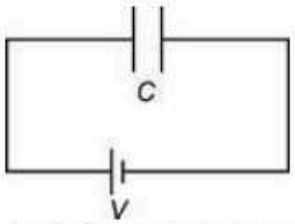
(1) Only Statement-1 and Statement-2 are correct.

(2) Only Statement-2 and Statement-3 are correct.

(3) Only Statement-1 and Statement-3 are correct.

(4) All statements are correct.

6.



An air filled parallel plate capacitor having capacitance, C is connected with a battery of voltage V. It has charge Q, electric field E and energy density u. Now, battery is disconnected from capacitor and a dielectric plate is fully inserted between plates. Match the following columns on the basis of above information.

	Column-I (Quantities)		Column-II (Their Values (finally))
A.	Charge (Q)	(P)	Decreases
B.	Capacitance (C)	(Q)	Increases
C.	Electric field (E)	(R)	Remain same
D.	Energy density (u)	(S)	Becomes zero

(1) A-R, B-Q, C-S, D-S

(2) A-R, B-Q, C-P, D-P

(3) A-S, B-Q, C-R, D-R

(4) A-S, B-Q, C-P, D-P

7. A system consists of 3 particles each of mass m located at points (0, 0)m; (2, 4) m and (4, 2) m. The coordinate of centre of mass are (in m)

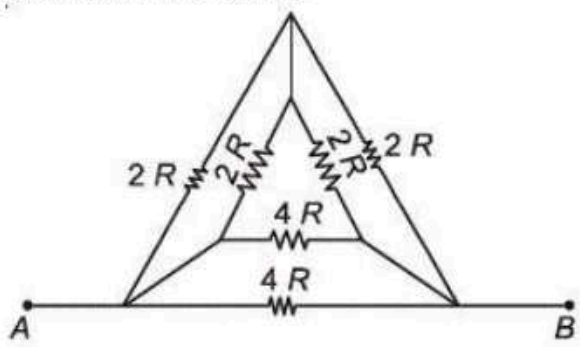
(1) (2, 2)

(2) (1, 1)

(3) (0, 0)

(4) $\left(\frac{1}{3}, \frac{1}{3}\right)$

8. The equivalent resistance across points A and B, in the circuit shown in the figure, is



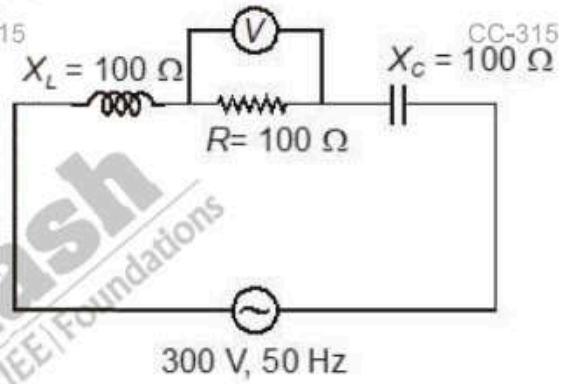
(1) 4 R

(2) 2 R

(3) 3 R

(4) R

9. In the circuit given below, the reading of the voltmeter V will be



(1) 300 V

(2) 150 V

(3) $\frac{150}{\sqrt{2}}$ V

(4) $300\sqrt{3}$ V

10. Match list-I with list-II.

	List-I		List-II
a.	Velocity	(i)	$[M^1L^1T^{-2}]$
b.	Acceleration	(ii)	$[M^0L^1T^{-2}]$
c.	Force	(iii)	$[M^1L^{-1}T^{-2}]$
d.	Pressure	(iv)	$[M^0L^1T^{-1}]$

Choose the correct answer from the options given below:

(1) a(iv), b(ii), c(i), d(iii)

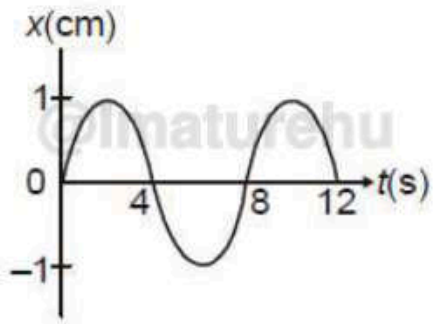
(2) a(iv), b(ii), c(iii), d(i)

(3) a(iii), b(ii), c(i), d(iv)

(4) a(ii), b(i), c(iii), d(iv)

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11. The x-t graph of a particle undergoing SHM is shown below. The acceleration of the particle at $t = \frac{4}{3}$ s is



(1) $\frac{\sqrt{3}}{32}\pi^3$ cm/s^2

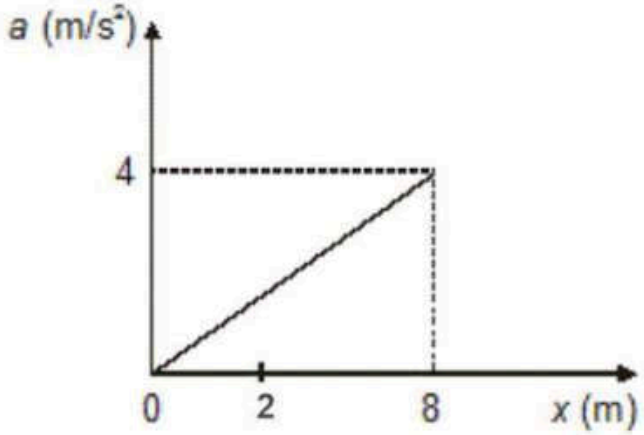
(2) $-\frac{\pi^2}{32}$ cm/s^2

(3) $\frac{\pi^2}{32}$ cm/s^2

(4) $-\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

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12. A 2 kg mass moving along *x*-axis whose acceleration-position graph is shown in figure. The work done on the particle from 2 m to 8 m is



(1) 45 J

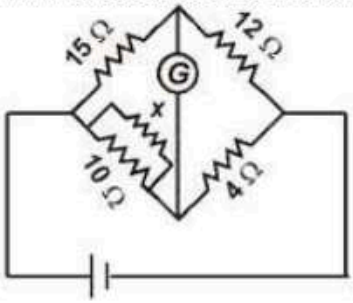
(2) Zero

(3) 30 J

(4) 15 J

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13. Four resistances of 15 Ω, 12 Ω, 4 Ω and 10 Ω respectively are connected in cyclic order to form Wheatstone bridge as shown in figure. The resistance *x* to be connected in parallel with the resistance of 10 Ω to balance the network is



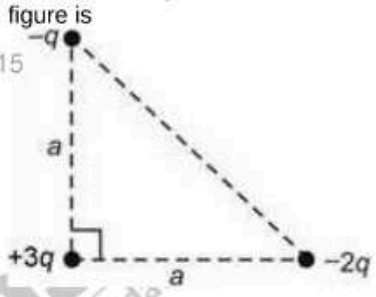
(1) 4 Ω

(2) 10 Ω

(3) 12 Ω

(4) 15 Ω

14. The electric dipole moment of the system as shown in the figure is



(1) qa

(2) Zero

(3) $\sqrt{5}qa$

(4) $\sqrt{2}qa$

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15. A parallel beam of light of wavelength 540 nm passes through a slit of width 0.2 mm. The transmitted light is collected on the screen 20 cm away. The distance between two first order minima is nearly

(1) 1 mm

(2) 2 mm

(3) 4 mm

(4) 3 mm

16. A body falls on a surface of coefficient of restitution 0.6 from a height of 1 m. Then the body rebounds to a height of

(1) 0.6 m

(2) 0.4 m

(3) 1 m

(4) 0.36 m

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17. A steel rod of length 1.0 m, area of cross-section 1 cm^2 is clamped between two rigid support. The tension developed in the rod on cooling the rod from 220°C to 20°C is (Y_S = 2 × 10^{11} N/m^2 and $\alpha_S = 10^{-5}$ /°C)

(1) 4×10^4 N

(2) 2×10^4 N

(3) 6×10^4 N

(4) Zero

18. Two lenses one diverging of power 3.0 D and other converging of power 1.0 D are combined together. The focal length of combination is

(1) 20 cm

(2) –20 cm

(3) 50 cm

(4) –50 cm

19. In uniform circular motion, which of the following quantities is zero?

(1) Change in kinetic energy

(2) Magnitude of change in velocity

(3) Change in magnitude of velocity

(4) Both (1) & (3)

20. If a body of mass *m* is raised to height 2 *R* from the earth's surface, then the change in potential energy of the body is (*R* is the radius of earth)

(1) *mgR*

(2) 12 *mgR*

(3) $\frac{2}{3}mgR$

(4) 9 *mgR*

21. The number of significant figures in all the measured value 25.12, 200.9, 4.156 and 1.217×10^{-4} is

(1) 1

(2) 2

(3) 3

(4) 4

22. An infinite number of charges, each of 2 μC are placed along x-axis at *x* = 1 m, 2 m, 4 m, 8 m ..... The total potential energy of 1 μC charge placed at the origin is

(1) 2.4×10^{-1} J

(2) 3.6×10^{-2} J

(3) 24 J

(4) 3.6 J

23. Let *r* be the distance of a point on the axis of a magnetic dipole from its centre. The magnetic field at such a point is proportional to (provided magnetic dipole is small)

(1) $\frac{1}{r}$

(2) $\frac{1}{r^2}$

(3) $\frac{1}{r^3}$

(4) $\frac{1}{r^0}$

24. When two waves of intensities I_1 and I_2 coming from coherent sources interfere at a point *P*, where phase difference is ϕ, then resultant intensity (I_{res}) at point *P* would be

(1) $I_{res} = I_1 - I_2 - 2\sqrt{I_1 I_2}$

(2) $I_{res} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

(3) $I_{res} = I_1 + I_2 - 2\sqrt{I_1 I_2}\cos^2\phi$

(4) $I_{res} = \sqrt{(I_1)^2 + (I_2)^2 - 2(I_1 I_2)\cos^2\phi}$

25. A ray of light is incident at an angle of incidence 40° on an equilateral prism and emerge at an angle of 40°, then the angle of deviation of the ray will be (in radian)

(1) $\frac{\pi}{9}$

(2) $\frac{\pi}{6}$

(3) $\frac{\pi}{3}$

(4) $\frac{2\pi}{3}$

26. A particle is moving in *x*-*y* plane, where *x*-axis is along the horizontal and *y*-axis is along vertically upward. Its co-ordinates are given as $x = 3t$, $y = 4t - 2t^2$, where *x* and *y* are in metre and *t* is in seconds. *x* co-ordinate of particle, when it is at maximum *y* co-ordinate, is

(1) 9 m

(2) 6 m

(3) 3 m

(4) 4 m

27. In the nuclear decay given below

$${}^A_Z X \rightarrow {}^{A}_{Z+1}Y \rightarrow {}^{A-4}_{Z-1}B* \rightarrow {}^{A-4}_{Z-1}B$$

the particles emitted in the sequence are

(1) β^-, α, γ

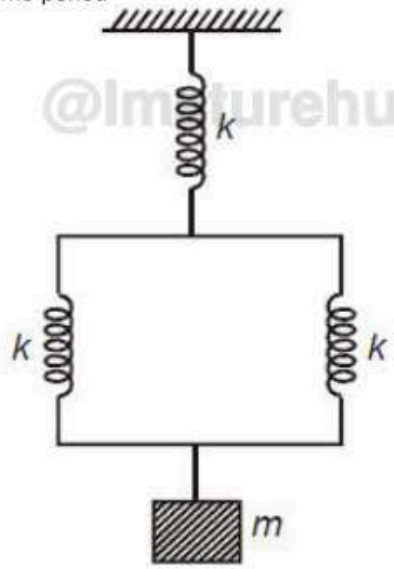
(2) γ, β^-, α

(3) α, β^-, γ

(4) β^-, γ, α

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28. A body of mass m hanging with the help of three springs, each of spring constant k as shown. If the mass is slightly displaced and released, then the system will oscillate with time period



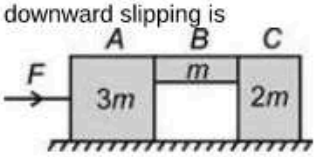
(1) $2\pi\sqrt{\frac{m}{3k}}$

(2) $2\pi\sqrt{\frac{2m}{3k}}$

(3) $2\pi\sqrt{\frac{3m}{2k}}$

(4) $2\pi\sqrt{\frac{3k}{m}}$

29. The system of three masses $3m$, m and $2m$ are pushed by a force F as shown in figure. All the surfaces are smooth except between B and C. Coefficient of friction between B and C is $\frac{1}{2}$. Minimum value of F to prevent block B from downward slipping is



(1) $3mg$

(2) $4mg$

(3) $5mg$

(4) $6mg$

30. A charged particle has acceleration $\vec{a} = 2\hat{i} + x\hat{j}$ in a magnetic field $\vec{B} = -3\hat{i} + 2\hat{j} - 4\hat{k}$. The value of x is

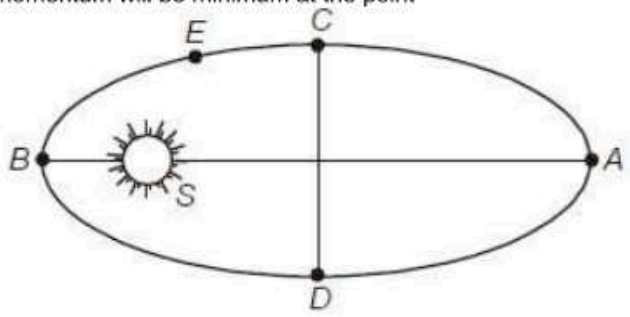
(1) 4

(2) 2

(3) 6

(4) 3

31. The earth E moves in an elliptical orbit with the sun S at one of the focii as shown in figure. The magnitude of its linear momentum will be minimum at the point



(1) C

(2) B

(3) A

(4) D

32. The energy of a hydrogen atom in the ground state is –13.6 eV. The energy of a He^+ ion in the 1st excited state will be

(1) –13.6 eV

(2) –27.2 eV

(3) –54.4 eV

(4) –6.8 eV

33. A thin liquid film formed between a U shaped wire and a light slider supports a weight of 1.5×10^{-2} N. The length of the slider is 30 cm and its weight negligible. The surface tension of the liquid film is



(1) 0.0125 Nm^{-1}

(2) 0.1 Nm^{-1}

(3) 0.05 Nm^{-1}

(4) 0.025 Nm^{-1}

34. **Assertion:** If light of some frequency $(v > v_0)$ is incident on a metal then all electron emit with same kinetic energy.
Reason: Kinetic energy of electrons is directly proportional to frequency of light in photoelectric emission.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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35. A body covers first one-third of the distance with a velocity 10 ms^{-1} in same direction, the second one-third with a velocity 20 ms^{-1} and last one-third with a velocity of 30 ms^{-1}. The average velocity of body is

(1) 17.8 ms^{-1}

(2) 16.4 ms^{-1}

(3) 18.3 ms^{-1}

(4) 20.2 ms^{-1}

36. A man running on a horizontal road at 6 km/h, finds the rain falling vertically. He doubles his speed and finds that the raindrops make an angle 37° with the vertical. The velocity of rain with respect to ground is

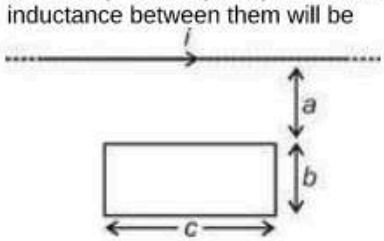
(1) 10 km/h

(2) 5 km/h

(3) 12 km/h

(4) 6 km/h

37. A rectangular coil is placed near a long wire carrying a current i (in same plane) as shown in the figure. The mutual inductance between them will be



(1) $\frac{\mu_0 a}{2\pi}\ln\left(\frac{c}{b}\right)$

(2) $\frac{\mu_0 C}{2\pi}\ln\left(1+\frac{b}{a}\right)$

(3) $\frac{\mu_0 a}{2\pi}\ln\left(1+\frac{b}{c}\right)$

(4) $\frac{\mu_0 c}{2\pi}\ln\left(\frac{a}{b}\right)$

38. Given below are two statements, one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): When a charge particle is placed in the cavity inside a uniformly charged non conducting sphere, then it experiences the same force at all positions inside the cavity.
Reason (R): Inside a cavity of uniformly charged sphere, the electric field is uniform.
In the light of above statements, choose the correct option.

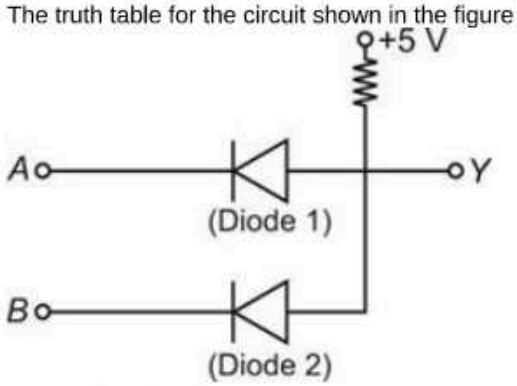
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

39. The truth table for the circuit shown in the figure is



(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

40. A torque of 50 Nm acting on a wheel at rest rotates it through 200 radians in 5 s. The angular acceleration produced by torque is

(1) 16 rad/s^2

(2) 12 rad/s^2

(3) 8 rad/s^2

(4) 4 rad/s^2

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41. In Young's double slit experiment, slit separation is *d* and distance of screen is *D*. Light beams incident from the slits are in phase. A glass slab of thickness '*t*' and refractive index 'μ' is introduced in path of one beam such that the central maxima shifts to position of 2nd minima. What is wavelength of light used.

(1) $2(\mu - 1)t$

(2) $\frac{2(\mu-1)t}{3}$

(3) $\frac{4(\mu-1)t}{3}$

(4) $\frac{(\mu-1)t}{2}$

42. The equation $y(x, t) = \frac{1.8}{(2x+5t)^2+5}$ represents a moving pulse, where *x* and *y* are in metre and *t* is in second. Then

(1) The pulse is travelling in positive *x* direction

(2) In 2 s it will travel a distance of 5 m

(3) The maximum displacement of the particle is 0.16 m

(4) In 2 s it will travel a distance of 1.25 m

43. Consider the magnetic field produced by a infinitely long current carrying wire. The correct option out of following is

(1) The lines of force will be concentric circles with centres on the wire

(2) There can be two points in the same plane where magnitude of magnetic fields are same

(3) There can be large number of points where the magnitude of magnetic field is same

(4) All of these

44. The speed of longitudinal wave in a gas depends on pressure as (At constant temperature)

(1) $v \propto P^{\frac{1}{2}}$

(2) $v \propto P^0$

(3) $v \propto P^{-1/2}$

(4) $v \propto P$

45. A Carnot engine working as a refrigerator between 250 K and 300 K receives 500 cal of heat from the cold reservoir. Heat rejected by it to the hot reservoir is

(1) 400 cal

(2) 450 cal

(3) 550 cal

(4) 600 cal

CHEMISTRY

46. If the mass ratio of He to N_2 in a container is 2 : 1 then the ratio of their moles will be

(1) 2 : 1

(2) 4 : 1

(3) 7 : 1

(4) 14 : 1

47. Mass of zinc required to produce 2.24 L of H_2 at STP on treatment with dilute sulphuric acid is (Atomic wt. of Zn = 65 u)

(1) 6.5 g

(2) 3.25 g

(3) 13.0 g

(4) 4.5 g

48. Given below are two statements:
Statement-I: For hydrogen atom n = 1, the electron has a more negative energy than n = 5, which means that the electron is more loosely bound in n = 1 orbit
Statement-II: Maximum number of electrons that a subshell can accommodate is $4l + 2$.
In the light of above statements, choose the correct answer from the options given below.

(1) Both statements I and II are correct

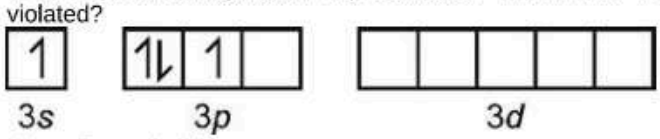
(2) Statement I is incorrect but statement II is correct

(3) Statement I is correct but statement II is incorrect

(4) Both statements I and II are incorrect

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49. In the given configuration representation which rule is violated?



(A) Aufbau principle
(B) Pauli's exclusion principle
(C) Hund's rule of maximum multiplicity
Choose the correct option.

(1) (A), (B) and (C)

(2) (A) and (C) only

(3) (B) and (C) only

(4) (B) only

50. Match List-I with List-II

	List-I Species		List-II Hybridisation and bond angle
a.	H_2O	(i)	sp^2 and 120°
b.	SO_2	(ii)	sp^3 and 104.5°
c.	NH_3	(iii)	sp^2 and 119.5°
d.	BF_3	(iv)	sp^3 and 107°

The correct match is

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(ii), b(iii), c(iv), d(i)

51. Among the following processes, in which process hybridization of underline atom does not change?

(1) $\underline{Al}Cl_3 + Cl^- \rightarrow \underline{Al}Cl_4^-$

(2) $\underline{N}H_3 + H^+ \rightarrow \underline{N}H_4^+$

(3) $\underline{C_2}H_4 + H_2 \rightarrow \underline{C_2}H_6$

(4) $\underline{S}O_3 + H_2O \rightarrow H_2\underline{S}O_4$

52. 2 moles of a gas is expanded isothermally and reversibly from 10 lit. to 50 lit at 300 K temperature. Work done during the process is (Given $\log_{10}5 = 0.7$)

(1) – 8.04 kJ

(2) + 12.5 kJ

(3) – 4.01 kJ

(4) + 6.2 kJ

53. The standard enthalpy of formation of $H_2O(l)$ and glucose (s) at $25^\circ C$ are $-300\ kJ\ mol^{-1}$ and $-1300\ kJ\ mol^{-1}$ respectively and standard enthalpy of combustion of C (graphite) is $-400\ kJ\ mol^{-1}$. The ΔU of combustion of glucose per gram at $25^\circ C$ will be

(1) $-3004\ kJ$

(2) $-2900\ kJ$

(3) $-16.7\ kJ$

(4) $-16.11\ kJ$

54. $2.5\ mL$ of $\frac{2}{5}M$ weak monoacidic base ($K_b = 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}M$ HCl in water at 25°C. The pH at equivalence point is

(1) 1.5

(2) 2.3

(3) 3.7

(4) 4.7

55. Consider the following reaction
$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g),\ \Delta H = -92.2\ kJ$
Choose the condition(s) from the following which favour(s) the formation of methanol.
(a) Decreasing the pressure.
(b) Addition of inert gas at constant volume
(c) Decreasing the temperature

(1) (a) only

(2) (a) and (b) only

(3) (c) only

(4) (a) and (c) only

56. Given below are two statements.
Statement (I): Oxidation state of terminal Br in Br_3O_8 is + 7.
Statement (II): Average oxidation state of Br in Br_3O_8 is $\frac{+16}{3}$.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement (I) is correct but statement (II) is incorrect

(2) Statement (I) is incorrect but statement (II) is correct

(3) Both statement (I) and statement (II) are correct

(4) Both statement (I) and statement (II) are incorrect

57. Identify the **correct** statements from the following.
(a) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(b) CO is highly poisonous in nature.
(c) CO_2 is used as fire extinguisher.
(d) The basic structural unit of silicates is SiO_4^{4-}.

(1) (a) and (b) only

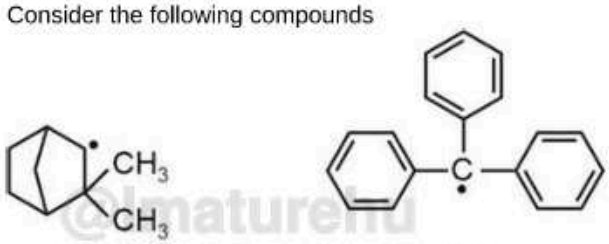
(2) (a) and (c) only

(3) (b) and (c) only

(4) (a), (b), (c) and (d)

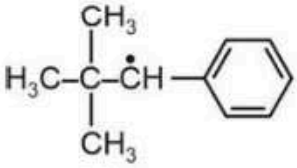
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58. Consider the following compounds

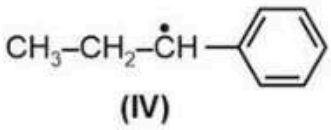


(I)

(II)



(III)



(IV)

Hyperconjugation does not occur in

(1) (I) only

(2) (I), (II) and (III) only

(3) (IV) only

(4) (I), (II), (III) and (IV)

59. How many isomers (including stereoisomers) of molecular formula $C_4H_{10}O$ will be possible?

(1) 8

(2) 9

(3) 7

(4) 10

60. In sulphur estimation 0.25 g of an organic compound gave 0.466 g of barium sulphate. Percentage of sulphur in the compound is (Molecular mass of $BaSO_4$ = 233 u)

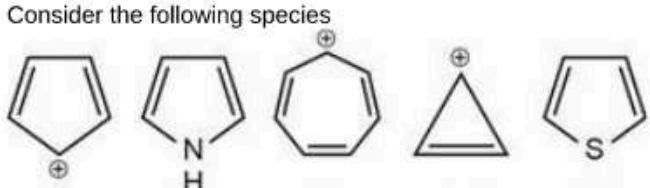
(1) 45.5%

(2) 16.4%

(3) 56.5%

(4) 25.6%

61. Consider the following species



(i) (ii) (iii) (iv) (v)

The total number of aromatic species are

(1) 2

(2) 4

(3) 5

(4) 3

62. Match List-I with List-II.

	List-I Reaction		List-II Catalyst
a.	$2\,CH_4 + O_2 \xrightarrow[100\ atm]{523\ K} 2\,CH_3OH$	(i)	Ni
b.	$CH_4 + H_2O \xrightarrow{1273\ K} CO + 3H_2$	(ii)	Cu
c.	$CH_4 + O_2 \xrightarrow{\Delta} HCHO + H_2O$	(iii)	$(CH_3COO)_2Mn$
d.	$2\,CH_3CH_3 + 3O_2 \xrightarrow{\Delta} 2\,CH_3COOH + 2H_2O$	(iv)	Mo_2O_3

Choose the correct answer from the given options.

(1) a-(i), b-(ii), c-(iv), d-(iii)

(2) a-(ii), b-(i), c-(iv), d-(iii)

(3) a-(ii), b-(i), c-(iii), d-(iv)

(4) a-(iv), b-(ii), c-(i), d-(iii)

63. 0.004 M solution of Na_2SO_4 is isotonic with a 0.0108 M solution of glucose at same temperature. The percentage dissociation of Na_2SO_4 is

(1) 75%

(2) 50%

(3) 25%

(4) 85%

64. Number of Faraday(s) required to get 1 mol of O_2 from electrolysis of water is

(1) 1

(2) 2

(3) 4

(4) 1.5

65. $Ni(s) \mid Ni^{2+}$ (1 M) || H^+(pH = x) | H_2 (1 bar) | Pt

The value of x for which the emf of the cell is zero at 25°C, will be [Given $E^o_{Ni\,|\,Ni^{2+}} = 0.236$ V]

(1) 4

(2) 6

(3) 3

(4) 2

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66. For the reaction, $3A + 2B + C \rightarrow$ Products, the following kinetic data was obtained

Initial concentration			
A(mol L^{-1})	B(mol L^{-1})	C(mol L^{-1})	Initial rate (mol L^{-1} s^{-1})
0.01	0.005	0.01	5×10^{-5}
0.02	0.005	0.01	5×10^{-5}
0.01	0.010	0.01	2.5×10^{-5}
0.01	0.005	0.02	20×10^{-5}

Hence, the rate of reaction is given by

(1) Rate = k [A] [B] [C]

(2) Rate = k $[A]^0$ $[B]^{-1}$ $[C]^2$

(3) Rate = k [A] [B] $[C]^{-2}$

(4) Rate = k [A] [B] $[C]^2$

67. If a 1st order reaction follows the given equation $\log k = \frac{-3000}{T} + 5$ where k is rate constant and T is temperature, then the value of pre-exponential factor 'A' and activation energy respectively are

(1) 10^5 and 57.44 kJ/mol

(2) 10^6 and 41.3 kJ/mol

(3) e^5 and 10 kJ/mol

(4) e^6 and 5 kJ/mol

68. Which of the following is a pair of peroxy compounds?

(1) H_2SO_5 and $H_2S_2O_8$

(2) $H_2S_4O_6$ and $H_2S_2O_8$

(3) H_2SO_5 and $H_2S_2O_3$

(4) $H_2S_4O_6$ and $H_2S_2O_3$

69. Correct order of HEH bond angle is (E = N, P, As, Sb)

(1) $SbH_3 > AsH_3 > PH_3 > NH_3$

(2) $NH_3 > PH_3 > AsH_3 > SbH_3$

(3) $NH_3 > PH_3 > SbH_3 > AsH_3$

(4) $PH_3 > NH_3 > AsH_3 > SbH_3$

70. The correct order of ionic radii is

(1) $Am^{3+} < Pu^{3+} < Np^{3+} < U^{3+} < Ac^{3+}$

(2) $Ac^{3+} < U^{3+} < Np^{3+} < Pu^{3+} < Am^{3+}$

(3) $U^{3+} < Ac^{3+} < Pu^{3+} < Np^{3+} < Am^{3+}$

(4) $Am^{3+} < Np^{3+} < Ac^{3+} < Pu^{3+} < U^{3+}$

71. Match list-I with list-II

	List-I (Octahedral complex)		**List-II (Moles of AgCl precipitated per mole of the compound with excess $AgNO_3$)**
a.	$CoCl_3 \cdot 4NH_3$	(i)	0
b.	$CoCl_3 \cdot 5NH_3$	(ii)	3
c.	$CoCl_3 \cdot 3NH_3$	(iii)	2
d.	$CoCl_3 \cdot 6NH_3$	(iv)	1

The correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(ii), c(iv), d(i)

72. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): cis-$[CrCl_2(ox)_2]^{3-}$ is a chiral complex anion.

Reason (R): cis-$[CrCl_2(ox)_2]^{3-}$ lacks plane of symmetry and centre of symmetry.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are false

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

73. The alkane that gives only one mono-chloro product on chlorination with Cl_2 in presence of diffused sunlight is

(1) Isopentane

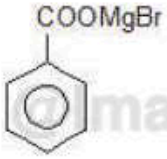
(2) 2-methylbutane

(3) 2, 2, 3, 3-tetramethyl butane

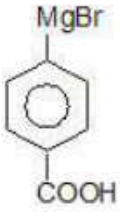
(4) n-pentane

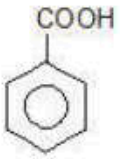
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74. Bromobenzene reacts with Mg in presence of ether to produce compound 'A' which reacts with $CO_2(g)$ to form compound 'B'. 'B' on hydrolysis produces

(1) 

(2) 

(3) 

(4) 

75. Match List-I with List-II.

List-I (Molecules)	List-II (Boiling point)
(a) Ethanol	(i) 390 K
(b) Methoxymethane	(ii) 231 K
(c) Propane	(iii) 248 K
(d) Butan-1-ol	(iv) 351 K

Choose the correct option.

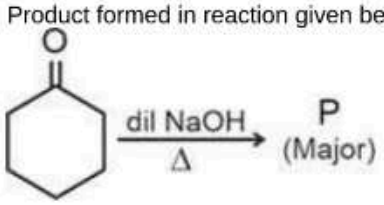
(1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

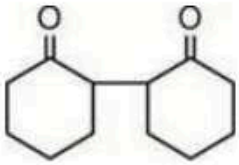
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

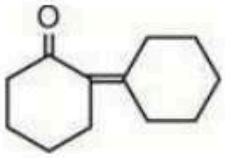
(3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

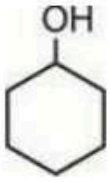
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

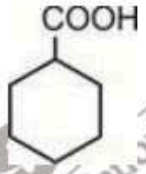
76. Product formed in reaction given below is,



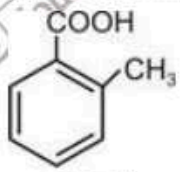
(1) 

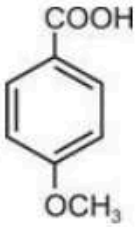
(2) 

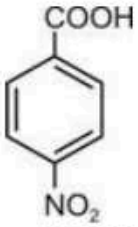
(3) 

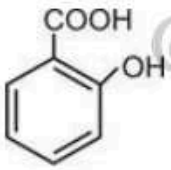
(4) 

77. The most acidic compound among the following is

(1) 

(2) 

(3) 

(4) 

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78. $C_6H_5NO_2 \xrightarrow[(ii)\ OH^-]{(i)\ Sn/HCl} A \xrightarrow[0-5^\circ C]{NaNO_2+HCl} B \xrightarrow[OH]{PhOH} C$ Product (C) is

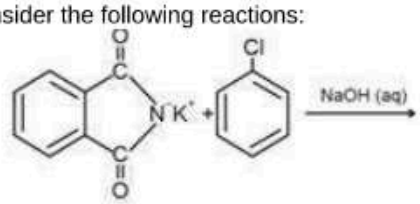
(1) p-Hydroxyazobenzene

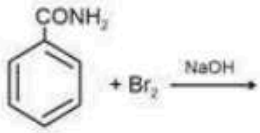
(2) p-Aminoazobenzene

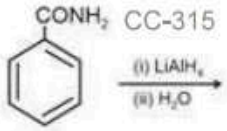
(3) p-Chloroazobenzene

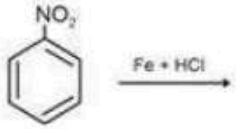
(4) p-Aminochlorobenzene

79. Consider the following reactions:

(A) 

(B) 

(C) 

(D) 

Identify the reaction(s) which give aniline as the major product.

(1) (A), (B) and (D) only

(2) (C) and (D) only

(3) (B) and (D) only

(4) (A), (B), (C) and (D)

80. Number of stereo centres present in linear structure of glucose is

(1) 4

(2) 3

(3) 5

(4) 6

81. Nessler's reagent is used to detect ammonia in water. Chemically it consists of

(1) $K_4[Fe(CN)_6]$

(2) K_2HgI_4

(3) $K_3[Fe(CN)_6]$

(4) KIO_3

82. Number of nodes in π^*2p_x, σ^*1s and $\pi 2p_x$ molecular orbitals respectively are

(1) 3, 1 and 2

(2) 2, 1 and 1

(3) 3, 2 and 1

(4) 2, 1 and zero

83. Match list-I with list-II.

	List-I (Molecule)		List-II (Shape)
a.	XeF_2	(i)	Bent T-shape
b.	ClF_3	(ii)	See-saw
c.	SF_4	(iii)	Pyramidal
d.	XeO_3	(iv)	linear

The correct match is

(1) a(iii), b(ii), c(i), d(iv)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(i), b(ii), c(iii), d(iv)

84. The hydrolysis products of lactose are

(1) α-D-Glucose and β-D-Fructose

(2) β-D-Galactose and β-D-Glucose

(3) β-D-Galactose and β-D-Fructose

(4) α-D-Glucose and β-D-Galactose

85. During the electrolysis of concentrated H_2SO_4, the reaction taking place at anode is

(1) $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$

(2) $2SO_4^{2-}(aq) \rightarrow S_2O_8^{2-}(aq) + 2e^-$

(3) $2SO_4^{2-}(aq) \rightarrow S_2O_7^{2-}(aq) + \frac{1}{2}O_2(g) + 2e^-$

(4) $2H_2O(l) \rightarrow 4H^+(aq) + O_2(g) + 4e^-$

86. Which of the following alcohols in Victor Meyer test gives red colour?

(1) Ethanol

(2) Propan-2-ol

(3) 2-methylpropan-2-ol

(4) Pentan-2-ol

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87. Consider the following statements:
(a) The correct electronic configuration of Berkelium(Bk) is $[Rn]5f^{9}7s^{2}$.
(b) Nickel complexes are useful in the polymerisation of alkynes.
(c) V_2O_5 catalyses the oxidation of SO_2 in the manufacture of sulphuric acid.
(d) Mischmetall consists of almost 95% lanthanoid metal and 5% iron and traces of S, C, Ca and Al.
The correct statements are

(1) (a), (b), (c) and (d)

(2) (a) and (c) only

(3) (a), (b) and (c) only

(4) (c) and (d) only

88. Match group of cation in column-I with their group reagent of qualitative analysis in column-II

	Column-I		Column-II
a.	Group-I	(i)	$NH_4OH + NH_4Cl$
b.	Group-II	(ii)	dilute HCl
c.	Group-III	(iii)	H_2S + dilute HCl
d.	Group-IV	(iv)	$H_2S + NH_4OH$

Correct match among the following is

(1) a(iii), b(ii), c(iv), d(i)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(iii), c(i), d(iv)

89. Consider the following statements
Statement I : The increase in electronegativities across a period is accompanied by an increase in metallic properties.
Statement II : In Pauling scale, an arbitrary value of 4.0 is assigned to electronegativity of fluorine.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are incorrect

90. Select the correct statement(s) among the following
a. Electron gain enthalpy of As is more negative than Cl
b. (+3) oxidation state are shown by Sb and Bi both
c. P is more electronegative than N

(1) a & c only

(2) b only

(3) a & b only

(4) b & c only

BIOLOGY

91. Refer to the various taxa of housefly classification and arrange them in **correct** sequence of taxonomic hierarchy starting from the lowest rank.
(i) Arthropoda (iv) Animalia
(ii) Diptera (v) Muscidae
(iii) *Musca* (vi) Insecta

(1) (ii) → (v) → (vi) → (iii) → (iv) → (i)

(2) (iii) → (v) → (vi) → (ii) → (i) → (iv)

(3) (iii) → (v) → (ii) → (vi) → (i) → (iv)

(4) (iii) → (v) → (iv) → (vi) → (i) → (ii)

92. Agents that are abnormally folded proteins cause all the given diseases, **except**

(1) Creutz feldt Jacob disease

(2) Bovine spongiform encephalopathy

(3) Mad cow disease

(4) Potato spindle tuber disease

93. Euglenoids have

(1) Two flagella of same size

(2) Lipid rich layer called pellicle

(3) Pigments identical to those present in higher plants

(4) Heterotrophic mode of nutrition only in the presence of sunlight

94. Read the following statements (A – D).
(A) Cyanobacteria are photosynthetic prokaryotes, being the most primitive organism to have oxygenic photosynthesis.
(B) Bacterial structure is very simple but they are very complex in their behaviour.
(C) Smallest wall-less monerans are mostly autotrophic.
(D) Archaebacteria differ from other bacteria in having a different cell wall structure which is responsible for their survival in extreme conditions.
Select the option for the **correct** set of statements.

(1) Only (A), (B) and (C)

(2) Only (B) and (D)

(3) Only (C) and (D)

(4) Only (A), (B) and (D)

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95. Match the following columns and select the **correct** option regarding bryophytes.

	Column I		Column II
(a)	Gametophyte	(i)	Haploid, antherozoid producing structure
(b)	Sporophyte	(ii)	Flask shaped haploid structure
(c)	Antheridium	(iii)	Produces gametes
(d)	Archegonium	(iv)	Non photosynthetic diploid body

(1) a(iii), b(iv), c(ii), d(i)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(i), d(ii)

96. Choose the **correct** ones w.r.t. the given plant.



(a) Many species of it are used as food.
(b) It majorly possesses chlorophyll *a*, *d* and fucoxanthin
(c) Cell wall of the given plant is made of cellulose and algin.
(d) Gametes are pyriform and bear two laterally attached flagella
(e) The commercial product obtained from the given plant is used to grow microbes and in preparations of jellies

(1) (a), (b) and (c) only

(2) (b), (d) and (e) only

(3) (c) and (d) only

(4) (a), (c) and (d) only

97. Main plant body is thalloid and has rhizoids in

(1) *Marchantia*

(2) *Selaginella*

(3) *Ginkgo*

(4) *Eucalyptus*

98. Identify the **correct** match w.r.t. parts of the root.
(1) Meristematic zone – Very small, thick walled cells
(2) Elongation zone – Cells divide repeatedly
(3) Maturation zone – Proximal to the region of elongation
(4) Root cap – Unicellular, thimble like structure

(1) (1)

(2) (2)

(3) (3)

(4) (4)

99. Consider the following floral features.
a. Persistent calyx
b. Placenta swollen with one ovule
c. Fruit pod or capsule
d. Albuminous seed
e. Ex-stipulate leaves
Which of the given characteristics are found in solanaceae?

(1) a, b, d, e

(2) a, d, e

(3) a, c, d, e

(4) b, d, e

100. Which of the following characteristics are **true** for China rose?
(i) Alternate phyllotaxy
(ii) Valvate aestivation of corolla
(iii) Superior ovary
(iv) Pentacarpellary gynoecium
(v) Polyadelphous androecium
(vi) Axile placentation
The **correct** ones are

(1) (i), (ii), (iv) and (v)

(2) (i), (iii), (iv) and (vi)

(3) (iii), (iv) and (v)

(4) (i), (ii) and (vi)

101. Conjoint type of vascular bundles

(1) Are common in roots

(2) Have the phloem located only on the inner side of xylem

(3) Have xylem & phloem jointly situated at the same radius

(4) Are only closed types

102. A student observed V.S. of some plant structure having following features.
(a) Conjoint but closed vascular bundles.
(b) Epidermis on upper and lower surfaces with conspicuous cuticle.
(c) Chloroplast containing parenchymatous tissue divided into two regions.
(d) Stomata mostly on abaxial surface.
Identify the plant structure based on the given features.

(1) Monocot stem

(2) Dicot root

(3) Dicot leaf

(4) Monocot leaf

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103. Read the following statements and choose the option with **correct** set of statements regarding plasma membrane.
a. Phospholipids have non-polar tail of saturated hydrocarbons.
b. Head part of the lipids is protected from the aqueous environment.
c. Transmembrane proteins require crude methods of extraction.
d. The quasi fluid nature of lipids is important for cell growth and secretion.
e. Lipids can do flip-flop movement.

(1) a, c and e only
(2) a, c and d only
(3) a, c, d and e only
(4) a, b, d and e only

104. Select the **correct** match

(1) Capsule – Allows bacterium to hide from host's immune system
(2) Ribosome – Non-membranous ribonucleoprotein particles found in all cells except few bacteria
(3) Vacuole – Polymorphic cell organelle that maintains osmotic pressure of cell
(4) Peroxisome – Involves in photorespiration but no role in animal cell

105. Colourless plastids that store starch is

(1) Elaioplasts
(2) Aleuroplasts
(3) Amyloplasts
(4) Chromoplasts

106. How many of the below given features belong to smooth muscle fibres?

Fusiform, Multinucleated, Involuntary, Presence of striations, Attached to skeletal bones

Choose the **correct** option.

(1) Two
(2) Three
(3) Four
(4) Five

107. **Assertion (A):** Maximum reabsorption in a nephron occurs in PCT.
Reason (R): Brush bordered cuboidal epithelium present in PCT increase the surface area of reabsorption of nutrients from glomerular filtrate.
In the light of above statements, select the correct option.

(1) Both Assertion & Reason are true and reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

108. Brood pouch in male cockroach is bounded by

(1) 9th and 10th terga dorsally and 9th sternum ventrally
(2) 9th tergum dorsally and 9th and 10th sterna ventrally
(3) 7th, 8th and 9th sterna
(4) 7th tergum dorsally and 7th, 8th and 9th sterna ventrally

109. Select the **incorrect** statement w.r.t. frogs.

(1) The buccal cavity leads to the oesophagus through pharynx.
(2) Oesophagus is a long tube that opens into the stomach
(3) The length of intestine is reduced.
(4) Pancreas is a digestive gland which produces digestive enzymes.

110. Which of the following feature has played a major role in the success of insects on land?

(1) Unjointed appendages
(2) Chitinous exoskeleton
(3) Body divisions
(4) Open circulatory system

111. Chondrichthyes differ from Osteichthyes as former possess

(1) Operculum
(2) Air bladder
(3) Placoid scales
(4) Two-chambered heart

112. Cloaca is present in
(a) All the birds
(b) Non-mammalian tetrapods
(c) All gnathostomes
(d) All vertebrates
Choose the **correct** option.

(1) (a) and (b) only
(2) (c) only
(3) (d) only
(4) (b) only

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113. Read the following statements w.r.t. lipids.
(a) Arachidonic acid has 20 carbon atoms including the carboxyl carbon.
(b) Fatty acids could be saturated or unsaturated.
(c) Many lipids are formed by both glycerol and fatty acids.
(d) Palmitic acid has 16 carbon atoms excluding the carboxyl carbon.
(e) When fatty acids are esterified with glycerol, they can be transformed into monoglycerides, diglycerides or triglycerides.
Choose the option with **incorrect** statement(s) only.

(1) (c) and (e)

(2) Only (d)

(3) (a) and (b)

(4) Only (e)

114. Coenzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin

(1) Thiamine

(2) Niacin

(3) Pyridoxine

(4) Tocopherol

115. Which of the following statements best explains how do biocatalyst accelerates the rate of chemical conversions?
[Note: S = Substrate and P = Product]

(1) They help to increase V_{max} of a chemical reaction

(2) They help to increase the temperature of reaction hence lower the energy level difference between S and P

(3) They bring down the difference in average energy content of 'S' from that of transition state, making the transition of 'S' to 'P' more easy

(4) They increase the activation energy of reaction hence, help in speeding up the conversion of 'S' into 'P'

116. Nucleolus is

(1) Single membrane bound structure

(2) Smaller in cells that are actively involved in protein synthesis.

(3) Site for ribosomal RNA synthesis.

(4) Separated from nucleoplasm by a membrane

117. In animal cell, cytokinesis

(1) Occurs via cell plate formation

(2) Is achieved by formation of furrow

(3) Progresses from centre to periphery

(4) Occurs centrifugally

118. Select the **correct** sequence of events that take place during meiosis.
(i) Reduction in the number of chromosomes.
(ii) The crossing over is completed.
(iii) The bivalent chromosomes align themselves on the equatorial plate.
(iv) Chromatids move to the opposite pole.

(1) (ii) $\longrightarrow$ (iii) $\longrightarrow$ (i) $\longrightarrow$ (iv)

(2) (ii) $\longrightarrow$ (iii) $\longrightarrow$ (iv) $\longrightarrow$ (i)

(3) (iii) $\longrightarrow$ (i) $\longrightarrow$ (ii) $\longrightarrow$ (iv)

(4) (iii) $\longrightarrow$ (ii) $\longrightarrow$ (i) $\longrightarrow$ (iv)

119. Match the given columns and select the **correct** option.

	Column I		Column II
(a)	Thickest and shortest chromosome	(i)	Prophase
(b)	Phase marks the end of cell division	(ii)	Metaphase
(c)	Chromosomes appear like ball of wool	(iii)	Quiescent stage
(d)	Cells remain metabolically active but do not proliferate	(iv)	Cytokinesis

(1) a-(ii) b-(iv) c-(i) d-(iii)

(2) a-(iv) b-(iii) c-(i) d-(ii)

(3) a-(ii) b-(iii) c-(i) d-(iv)

(4) a-(i) b-(ii) c-(iii) d-(iv)

120. Select the **incorrect** statements from the following.
(a) Hatch and Slack pathway occurs in mesophyll cells of all types of plants.
(b) ATP required for phosphorylation during reduction stage in Calvin cycle comes from mitochondria.
(c) In *Sorghum*, the primary CO_2 acceptor is a 3-C molecule.
(d) C_4 plants lack photorespiration.

(1) (a) & (d) only

(2) (a) & (b) only

(3) (b) & (c) only

(4) (a), (b) & (c) only

121. Select the **correct** statement w.r.t. Calvin cycle.

(1) Carboxylation of RuBP results in the formation of 2 molecules of oxaloacetic acid

(2) Regeneration of RuBP is crucial for Calvin cycle for uninterrupted continuation of cycle

(3) The regeneration of one RuBP molecule required four ATP for phosphorylation

(4) To make one molecule of glucose, 3 turns of Calvin cycle are required

122. In respiration-mediated breakdown of carbohydrates and proteins, which of the following biomolecules is common?

(1) Dihydroxy acetone phosphate

(2) Glucose-6-phosphate

(3) Glyceraldehyde-3-phosphate

(4) Pyruvic acid

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123. Oxidative phosphorylation is

(1) Formation of ATP by transfer of phosphate group from a substrate to ADP

(2) Oxidation of phosphate group in ATP

(3) Addition of phosphate group to ATP

(4) Formation of ATP by energy released from electrons removed during substrate oxidation

124. Match the Column-I and Column-II w.r.t. phytohormones and their functions.

	Column-I		Column-II
(A)	Auxin	(i)	Increases size of apple
(B)	Abscisic acid	(ii)	Controls the xylem differentiation
(C)	Ethylene	(iii)	Closure of stomata
(D)	Gibberellin	(iv)	Fruit ripening

	A	B	C	D
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iv)	(ii)	(i)	(iii)
(4)	(iii)	(i)	(ii)	(iv)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

125. Leaves of rice seedling remain out of water due to

(1) Increase in indole compound that leads to rapid division of the cells

(2) Increased ABA concentration that increases the length of the stem.

(3) Gaseous phytohormone that induces rapid elongation of internode and petiole.

(4) Increased concentration of cytokinin that inhibits the growth of lateral bud

126. The embryo sac of angiosperms represents

(1) Male gamete

(2) Female gamete

(3) Male gametophyte

(4) Female gametophyte

127. Outbreeding device which prevents autogamy but not geitonogamy is

(1) Being monoecious as in castor

(2) Being dioecious as in papaya

(3) Self incompatibility as in tobacco

(4) Bisexuality as in *Hibiscus*

128. Read the following statements and mark them as **True** (T) or **False** (F).

a. In coconut, endosperm is its edible part.

b. Both endosperm and perisperm are similar in ploidy level and their origin.

c. Coleoptile is a hollow foliar structure that encloses shoot apex and few leaf primordia.

d. Single, large shield shaped cotyledon of wheat embryo is known as epiblast.

	a	b	c	d
(1)	T	F	F	F
(2)	T	F	T	F
(3)	F	F	F	T
(4)	T	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

129. Read the given statements.

(A) In Down's syndrome and Turner's syndrome, the affected individuals are short statured and these are the results of aneuploidy.

(B) In the Morgan's experiment on *Drosophila,* when white eyed females were crossed with red eyed males then all the males in progeny were white eyed while all the females were red eyed.

(C) The phenotypic ratio of test cross is 1 : 1 in case of completely linked genes in dihybrid cross.

(D) When a cross is done between two pink flowered snapdragon plants then 50% of the progeny plant will be pink flowered.

The **correct** ones are

(1) Only (B), (C) and (D)

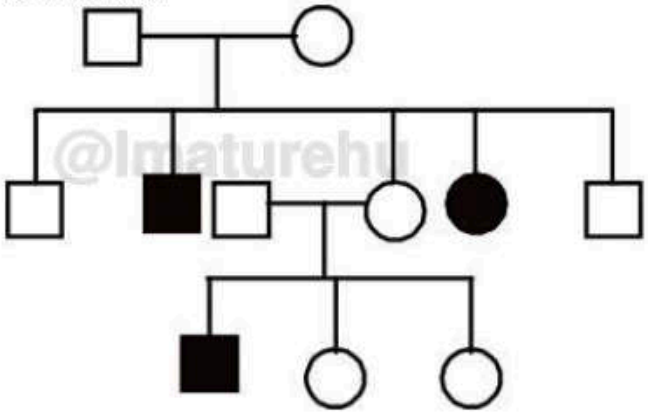
(2) Only (C) and (D)

(3) All (A), (B), (C) and (D)

(4) All except (D)

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130. Given below is a pedigree chart showing the inheritance of a trait in humans.



The trait traced in the given pedigree chart can be

(1) Myotonic dystrophy

(2) Sickle cell anaemia

(3) Haemophilia

(4) Colour blindness

131. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Asthma	(i)	Silica dust
b.	Emphysema	(ii)	Damage to alveolar walls
c.	Occupational respiratory disorder	(iii)	Hypersensitivity reaction

(1) a(i), b(ii), c(iii)

(2) a(iii), b(i), c(ii)

(3) a(ii), b(i), c(iii)

(4) a(iii), b(ii), c(i)

132. Read the following statements w.r.t. respiratory system in humans.

(A) At the tissue site, carbonic acid dissociates into H_2O and CO_2 in presence of enzyme carbonic anhydrase.

(B) Diffusion membrane is made up of three major cellular layers.

Choose the **correct** option.

(1) Both statements A and B are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

133. Match column I with column II and select the **correct** option.

Column I	Column II
(a) Angina	(i) Heart stops beating
(b) Cardiac arrest	(ii) Heart muscle is suddenly damaged by an inadequate blood supply
(c) Atherosclerosis	(iii) Acute chest pain
(d) Heart attack	(iv) Lumen of arteries becomes narrower due to deposits of calcium, fat, cholesterol, *etc*.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iii), b(iv), c(i), d(ii)

134. Closed circulatory system with 4 chambered heart is found in all of the following, **except**

(1) Birds

(2) Mammals

(3) Amphibians

(4) Crocodiles

135. The nodal structure located in the lower left corner of right atrium close to atrio-ventricular septum is

(1) SA node

(2) AV node

(3) AV bundle

(4) Purkinje fibres

136. Choose the **incorrect** statement w.r.t. excretory system

(1) Skin, lungs and liver also assist in excretion.

(2) Nearly 99% of filtrate is reabsorbed through different parts of nephron

(3) Renal calculi are insoluble masses of crystallised salts called oxalates

(4) Nitrogenous wastes like urea are added to dialysing fluid to maintain a concentration gradient.

137. Presence of glucose and ketone bodies in urine are indicative of

(1) Renal calculi

(2) Uremia

(3) Diabetes mellitus

(4) Diabetes insipidus

138. Amount of CO_2 removed by lungs in one hour is

(1) 200 mL

(2) 180 L

(3) 12 L

(4) 2000 mL

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139. Match column I with column II w.r.t diseases and their description.

Column I		Column II	
a.	Gout	(i)	Auto-immune disorder affecting the neuro-muscular junction
b.	Tetanus	(ii)	Lack of relaxation between successive stimuli in sustained muscle contraction
c.	Osteoporosis	(iii)	Age-related disorder characterised by decreased bone mass
d.	Myasthenia gravis	(iv)	Inflammation of joints due to accumulation of uric acid crystals

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iii), b(iv), c(i), d(ii)

140. In humans, the part of myofilament which has binding sites for ATP and active sites for thin filaments is

(1) Head of heavy meromyosin

(2) Short arm of light meromyosin

(3) Tail part of light meromyosin

(4) Short arm of heavy meromyosin

141. When a neuron is not conducting any impulse *i.e.* it is at resting stage, the axonal membrane show all of the following events, **except**

(1) Permeability for K^+

(2) Nearly impermeable for Na^+

(3) Impermeability for negatively charged proteins

(4) Active transport of $2Na^+$ outside for $3K^+$ into the cell

142. Match column I with column II w.r.t. different types of neurons and their respective location/presence in humans.

Column I		Column II	
(a)	Multipolar neuron	(i)	Retina
(b)	Bipolar neuron	(ii)	Embryonic stage
(c)	Unipolar neuron	(iii)	Cerebral cortex

Select the correct option.

(1) (a)(i), (b)(ii), (c)(iii)

(2) (a)(iii), (b)(i), (c)(ii)

(3) (a)(iii), (b)(ii), (c)(i)

(4) (a)(ii), (b)(i), (c)(iii)

143. Select the **correct** match.

(1)	Addison's disease	–	Hypersecretion of adrenal cortex hormones
(2)	Graves' disease	–	Hyperthyroidism
(3)	Diabetes mellitus	–	Prolonged hypoglycemia
(4)	Acromegaly	–	Hyperproduction of growth hormone in children

(1) (1)

(2) (2)

(3) (3)

(4) (4)

144. Read the following statements.

a. It maintains cardio-vascular system and kidney functions.

b. It is given to patients who have undergone transplantation to suppress their immune response.

c. It is the main glucocorticoid which also stimulates RBCs production.

Select the option representing the hormone for which all of the above statements are true.

(1) Cortisol

(2) Aldosterone

(3) Epinephrine

(4) Thyroxine

145. In males, _____ stimulates the synthesis and secretion of androgens. In females, _____ stimulates the growth and development of ovarian follicles.

Choose the option that **correctly** fill the blanks respectively.

(1) LH, FSH

(2) GnRH, prolactin

(3) Testosterone, progesterone

(4) ADH, relaxin

146. Select the term used for cushion of fatty tissue present above labia majora and is covered by skin and pubic hairs

(1) Mons pubis

(2) Labia minora

(3) Clitoris

(4) Hymen

147. Signals for parturition originate from _____ developed foetus and _____in human females.

Choose the option which fills the respective blanks with **correct** words.

(1) Half, placenta

(2) Fully, placenta

(3) Half, mammary glands

(4) Fully, mammary glands

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148. Which path is followed by milk during milk ejection reflex?
(a) Mammary duct
(b) Mammary tubule
(c) Mammary ampulla
(d) Lactiferous duct
(e) Mammary alveoli

(1) e → b → a → c → d
(2) e → a → b → c → d
(3) e → b → d → c → a
(4) e → a → c → b → d

149. In human females, first meiotic division starts within

(1) Gamete mother cell
(2) Primary oocyte
(3) Tertiary follicle
(4) Secondary oocyte

150. Read the following statements and choose the option which identifies them as true (T) or false (F) correctly.
(a) Menarche is the first menstruation in women.
(b) Sperm formation may continue till old age in human males.
(c) Absence of hymen always indicates loss of virginity.
(d) Oogenesis begins at puberty in human females.
Choose the **correct** option

(1) **a**–T, **b**–F, **c**–T, **d**–T
(2) **a**–F, **b**–T, **c**–F, **d**–F
(3) **a**–T, **b**–T, **c**–F, **d**–F
(4) **a**–T, **b**–F, **c**–T, **d**–F

151. Mark the event which is **not** associated with the intake of oral steroidal contraceptive pills to prevent pregnancy in humans.

(1) They inhibit secretion of FSH and LH
(2) They prevent implantation
(3) They alter the quality of cervical mucus which retards the entry of sperms
(4) They increase the thickness of myometrium and prevent their contraction

152. The contraceptive device inserted by the doctors or expert nurses in the uterus through vagina is

(1) Implant
(2) IUD
(3) Cervical cap
(4) Diaphragm

153. The geological time scale is divided into eras, periods and epochs. Under which of the following era, Silurian period falls?

(1) Proterozoic
(2) Paleozoic
(3) Mesozoic
(4) Coenozoic

154. Which of the following statement is **incorrect**?

(1) About 15 mya, *Ramapithecus* and *Dryopithecus* were existing.
(2) *Ramapithecus* were more like man
(3) The first human like being the hominid was *Homo habilis*
(4) *Homo erectus* used animal hides to protect their body and buried their dead ones

155. Complete the analogy
Bobcat : Tasmanian tigercat :: Lemur : _____

(1) Numbat
(2) Mole
(3) Spotted cuscus
(4) Tasmanian wolf

156. Match the content of column I with content of column II and choose the **correct** option.

	Column I		Column II
A.	Edward Wilson	(i).	His estimate placed global species diversity at about 7 million
B.	Robert May	(ii).	Showed that increased diversity contributed to higher productivity
C.	Alexander von Humboldt	(iii).	Described the combined diversity at all levels of biological organisation
D.	David Tilman	(iv).	Observed that within the region species richness increases with increasing explored area but up to a limit

(1) A(ii), B(iv), C(i), D(iii)
(2) A(iv), B(i), C(ii), D(iii)
(3) A(iii), B(i), C(iv), D(ii)
(4) A(iii), B(iv), C(i), D(ii)

157. The disease sickle cell anaemia can be transmitted from parent to offspring when

(1) Only mother is the carrier for the gene
(2) Only father is the carrier for the gene
(3) Both the parents are the carrier for the gene
(4) Only father carries homozygous recessive trait

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158. Experimental verification of the chromosomal theory of inheritance was done by

(1) Sutton and Boveri

(2) T.H. Morgan

(3) Punnett

(4) Mendel

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159. Regarding Watson and Crick, the pitch of the DNA helix is

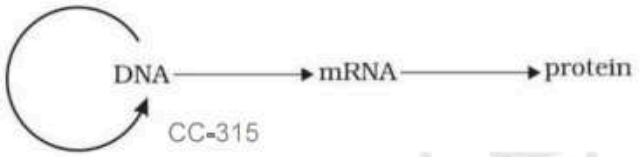
(1) 34 nm

(2) 0.34 nm

(3) 0.0034 nm

(4) 3.4 nm

160. Observe the following central dogma of molecular biology and the events occur during this.



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(a) Removal of introns and joining of exons in a defined order to form mRNA that is capable of translation.
(b) Polymerisation of deoxyribonucleotides
(c) Joining of discontinuously synthesised fragments by DNA ligase
(d) Formation of a replication fork
(e) Binding of RNA polymerase with promoter region of transcription unit
The **correct** sequential order of these events is:

(1) (d) → (b) → (e) → (c) → (a)

(2) (a) → (d) → (b) → (c) → (e)

(3) (a) → (b) → (c) → (e) → (d)

(4) (d) →(b) → (c) → (e) → (a)

161. "Genetic code is said to be degenerate." Which of the below given options elaborates this statement with suitable example.

(1) AUG acts as start codon and codes for methionine.

(2) Serine and leucine are coded by 6 codons.

(3) UUU codes for phenylalanine in all organisms except mitochondria.

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(4) Third nitrogenous base of a codon is usually not read by t-RNA.

162. How many amino acid sequences will be present in a polypeptide chain when genetic information flows from the following stretch of DNA?
3′ TACCATGC ATGCATC 5' Template strand
5′ ATGGTACG TACGTAG 3' coding strand

(1) 5

(2) 4

(3) 8

(4) 10

163. Select the **odd** one w.r.t. biological control of pests and disease

(1) *Trichoderma* is also used as biocontrol agent

(2) Some viruses can be used as biocontrol agents

(3) *Bacillus thuringiensis* is used to control caterpillar of butterfly

(4) Certain chemicals are used to kill the pests and to control diseases

164. Match column-I with column-II and choose the **correct** option.

Column-I	**Column-II**
a. *Trichoderma polysporum*	(i) Blood-cholesterol lowering agent
b. *Monascus purpureus*	(ii) Clot buster
c. *Streptococcus*	(iii) Immunosuppressive agent
d. *Acetobacter aceti*	(iv) Preparation of vinegar

(1) a(iii), b(ii), c(i), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iv), c(iii), d(ii)

165. Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.
Assertion (A) : In interference competition, even if resources are abundant, the feeding efficiency of both the species increases due to inhibitory presence of the other species.
Reason (R) : Resources must be limiting in any habitat for competition to occur.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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166. Which of the following are **correct** examples of population interactions, which show, '+' '–' pattern?
(a) Cuckoo laying its eggs in a crow nest
(b) *Penicillium* inhibiting growth of surrounding bacteria
(c) *Pisaster* and invertebrates in the American Pacific Coast
(d) Flamingoes and zooplanktons
(e) Abingdon tortoise and goats
Select the **correct** ones

(1) a, c and e only

(2) b, d and e only

(3) a, c and d only

(4) b, d and c only

167. Producers in the following food chain converts the energy from sun into the biomass of 100000 J energy. What would be amount of energy that will pass from 2nd to 3rd trophic level?
Plants → Caterpillar → Sparrow → Hawk

(1) 1000 J

(2) 100 J

(3) 10 J

(4) 10000 J

168. Find the **wrong** one w.r.t. productivity.

(1) GPP is the available biomass for consumption to heterotrophs.

(2) Annual NPP of whole biosphere is approximately 170 billion tons of organic matter.

(3) Deserts and deep sea are the least productive ecosystems.

(4) GPP is rate of production of organic matter by producers per unit area and time.

169. Choose the **incorrect** match w.r.t. biodiversity of tropical Amazonian rainforest in South America.

(1) Plant species — More than 40,000

(2) Reptiles species — 378

(3) Amphibians species — 427

(4) Birds species — 3000

170. In an ecosystem, rich species diversity is important to provide all of the following, **except**

(1) Resistance to invasions by alien species

(2) Resistance to occasional natural disturbances

(3) Resilience to man-made disturbances

(4) Much variations in productivity from year to year

171. *Streptococcus pneumoniae* and *Haemophilus influenzae* are the causative agents of

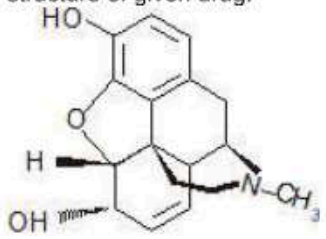
(1) Common cold

(2) Dysentry

(3) Pneumonia

(4) Diptheria

172. Choose the option that **correctly** identifies the chemical structure of given drug.



(1) Morphine

(2) Marijuana

(3) Hashish

(4) Cocaine

173. The yellowish colostrum secreted by the mother in the initial days of lactation is a rich source of IgA for the infant. These antibodies provide

(1) Natural active immunity to the infant

(2) Artificial passive immunity to the foetus

(3) Natural passive immunity to the infant

(4) Artificial active immunity to the infant

174. Restriction enzymes are also known as

(1) Molecular glue

(2) Molecular scissors

(3) Ligating agents

(4) Polymerising agents

175. Downstream processing of a desired product does **not** involve

(1) Packaging of product

(2) Purification of expressed protein

(3) Marketing of extracted product

(4) Expression of foreign gene

176. Method of introducing gene of interest/ foreign DNA directly into the nucleus of preferably an animal cell is

(1) Heat shock method

(2) Biolistics

(3) Micro-injection

(4) Gene gun

177. The two enzymes responsible for restricting the growth of bacteriophage in *E.coli* were isolated in year

(1) 1963

(2) 1983

(3) 1990

(4) 1997

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178. The enzyme that is used to degrade the bacterial cell wall is

(1) Pectinase

(2) Chitinase

(3) Cellulase

(4) Lysozyme

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179. All of the following characters are related with Bt toxin **except**

(1) Kills lepidopteran, dipteran and coleopteran insects

(2) The toxin is coded by a gene named *cry*

(3) Activated in midgut of insects in acidic medium

(4) Causes lysis of cells in epithelial lining of midgut

180. Which type of bonds were created between chains A and B to form human insulin?

(1) Ionic bonds

(2) Disulphide bonds

(3) Ester bonds

(4) Hydrogen bonds

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